UNIT – I - Laser System

UNIT - 1 Laser System

Basic Principle of Laser – Einstein Coefficients – Condition for light amplification – Population Inversion – Threshold Condition – Line Shape Function – Optical Resonators – Three level and four level systems.

I Introduction

LASER stands for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation. Laser is a device which emits a powerful, monochromatic collimated beam of light. The emitted light waves are coherent in nature.

The first laser, ruby laser was invented by Dr.T.H. Maiman in the year 1960. Since then, the development of lasers is extremely rapid. The laser action is being demonstrated in many solids, liquids, gases and semiconductor.

1.1 CHARACTERISTICS OF LASER

Laser is basically a light source. Laser light has the following important characteristics

- High Directionality
- High Intensity
- Highly Monochromatic
- Highly Coherence

1. Directionality

Ordinary light spreads in all directions and its angular spread is 1m/m.

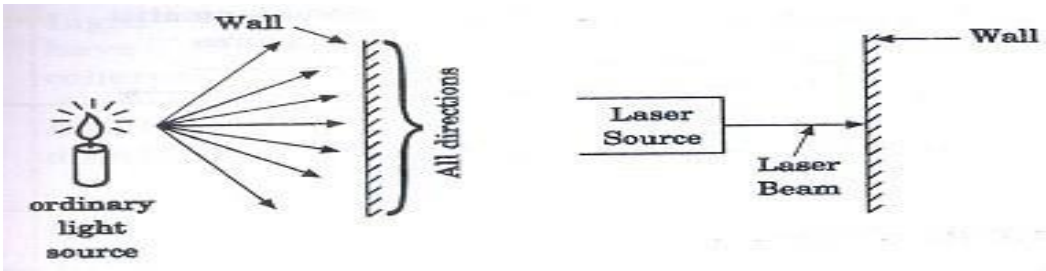


Fig. 1.1 Directionality property of Laser

But it is found that laser is highly directional and is angular spread is 1mm/m. For example, the laser beam can be focused to very long distance with a few divergence or angular spread shown in Fig. 1.1.

2. Intensity

Since an ordinary light spreads in all directions, the intensity reaching the target is very less.

But in the case of laser, due to high directionality, the intensity of laser beam reaching the target is of high intense beam. For example, 1 mill watt power of He-Ne laser appears to be brighter than the sunlight (Fig. 1.2).

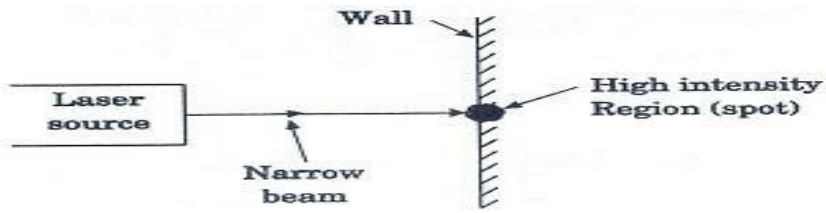


Fig. 1.2 Intensity variation

3. Monochromatic

Laser beam is highly monochromatic; the wavelength is single, whereas in ordinary light like mercury vapour lamp, many wavelengths of light are emitted Fig.1.3.

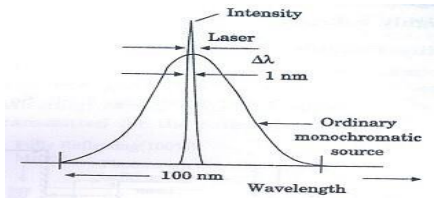


Fig. 1.3 Monochromaticity nature of Laser

4. Coherence

It is an important characteristic of laser beam. In lasers the wave trains of same frequency are in phase, the radiation given out is in mutual agreement not only in phase but also in the direction of emission and polarization. Thus it is a coherent beam. Due to high coherence it results in an extremely high power. Fig. 1.4 shows the coherence nature of Laser

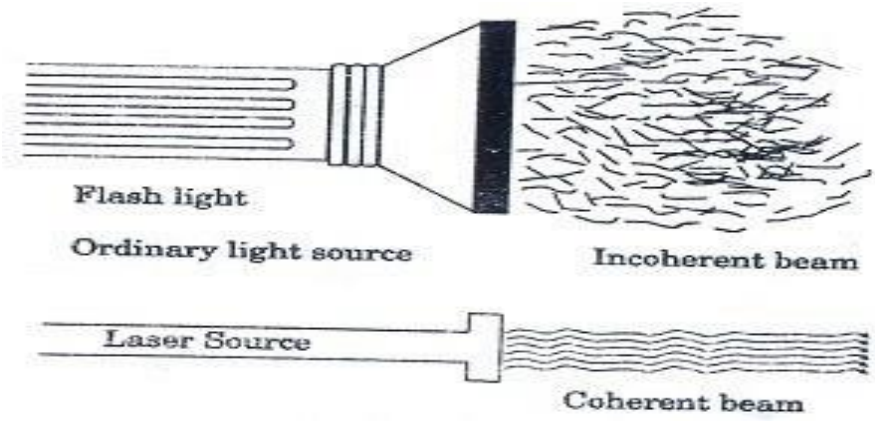


Fig. 1.4 shows the coherence property of Laser

Differences between ordinary light and Laser beam.

S.No.	**Ordinary light**	**Laser beams**
1	In ordinary light the angular spread is more	In laser beam the angular spread is less.
2	They are not directional.	They are highly directional.
3	It is less intense	It is highly intense
4	It is not a coherent beam and is not in	It is a coherent beam and is in phase

	phase.	
5	The radiation are polychromatic	The radiations are monochromatic
6	Example: Sun light, Mercury vapor lamp	He- Ne Laser, Co2 laser

1.2 STIMULATED ABSORPTION, SPONTANEOUS EMISSION AND STIMULATED EMISSION

Process 1 - Stimulated absorption

An atom in the lower energy level or ground state energy level E_1 absorbs the incident photon radiation of energy $h\nu$ and goes to the higher energy level or excited level E_2 as shown in figure 1.5. This process is called absorption.

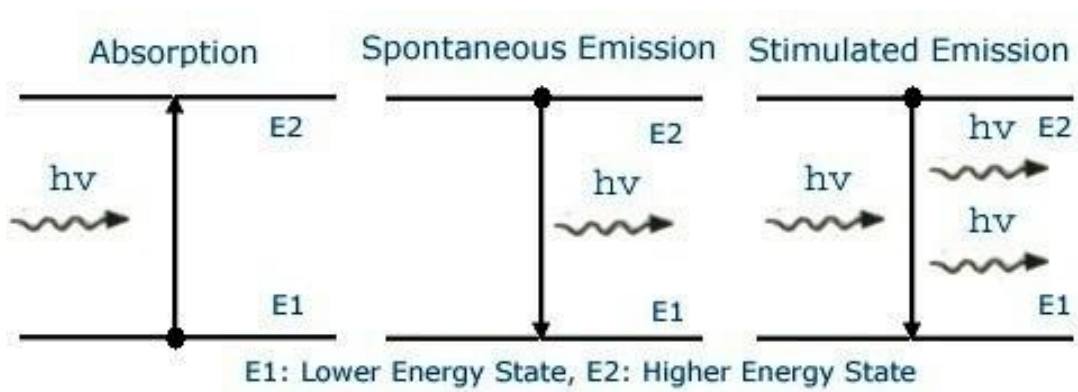


Fig. 1.5 Absorption and emission process in Laser

Process 2- Spontaneous Emission

The atom in the excited state returns to the ground state by emitting a photon of energy $E = (E_2 - E_1) = h\nu$ spontaneously without any external triggering as shown in the figure. This process is known as spontaneous emission. Such an emission is random and is independent of incident radiation.

Process 3 - Stimulated Emission

The atom in the excited state can also return to the ground state by external triggering or inducement of photon thereby emitting a photon of energy equal to the energy of the incident photon, known as stimulated emission. Thus results in two photons of same energy, phase difference and of same directionality as shown.

Table 1.1 Differences between Stimulated and spontaneous emission of radiation

S. No.	Stimulated Emission	Spontaneous emission
1.	An atom in the excited state is induced to return to the ground state , thereby resulting in two photons of same frequency and energy is called Stimulated emission	The atom in the excited state returns to the ground state thereby emitting a photon, without any external inducement is called Spontaneous emission.
2	The emitted photons move in the same direction and is highly directional	The emitted photons move in all directions and are random
3	The radiation is highly intense, monochromatic and coherent	The radiation is less intense and is incoherent
4	The photons are in phase, there is a constant phase difference	The photons are not in phase (i.e.) there is no phase relationship between them.
5	The rate of transition is given by $R_{12}(Sp) = A_{21} N_2$	The rate of transition is given by $R_{12}(Sp) = A_{21} N_2$

1.3 POPULATION INVERSION

Population Inversion creates a situation in which the number of atoms in higher energy state is more than that in the lower energy state. Usually at thermal equilibrium, the number of atoms N_2 i.e., the population of atoms at higher energy state is much lesser than the population of the atoms at lower energy state N_1 that is $N_1 > N_2$. The Phenomenon of making $N_2 > N_1$ is known as Population Inversion (Fig. 1.6).

Fig. 1.6 . Population Inversion

Condition for Population inversion

1. There must be at least two energy levels $E_2 > E_1$.
2. There must be a source to supply the energy to the medium.
3. The atoms must be continuously raised to the excited state.

1.4 META STABLE STATES

An atom can be excited to a higher level by supplying energy to it. Normally, excited atoms have short life times and release their energy in a matter of nano seconds (10^{-9}) through

spontaneous emission. It means atoms do not stay long to be stimulated. As a result, they undergo spontaneous emission and rapidly return to the ground level; thereby population inversion could not be established. In order to do so, the excited atoms are required to 'wait' at the upper energy level till a large number of atoms accumulate at that level. In other words, it is necessary that excited state have a longer lifetime.

A Meta stable state is such a state. Metastable can be readily obtained in a crystal system containing impurity atoms. These levels lie in the forbidden gap of the host crystal. There could be no population inversion and hence no laser action, if metastable states don't exist.

1.5 EINSTEIN'S "A & B" COEFFICIENTS - DERIVATION

We know that, when light is absorbed by the atoms or molecules, then it goes from the lower energy level (E_1) to the higher energy level (E_2) and during the transition from higher energy level (E_2) to lower energy level (E_1) the light is emitted from the atoms or molecules. Fig. 1.7., process involved in Laser.

Let us consider an atom exposed to light photons of energy, three distinct processes take place. ***(a). Absorption (b). Spontaneous emission (c). Stimulated Emission***

a). Absorption

An atom in the lower energy level or ground state energy level E_1 absorbs the incident photon radiation of energy $h\nu$ and goes to the higher energy level or excited level E_2 as shown in figure. This process is called absorption.

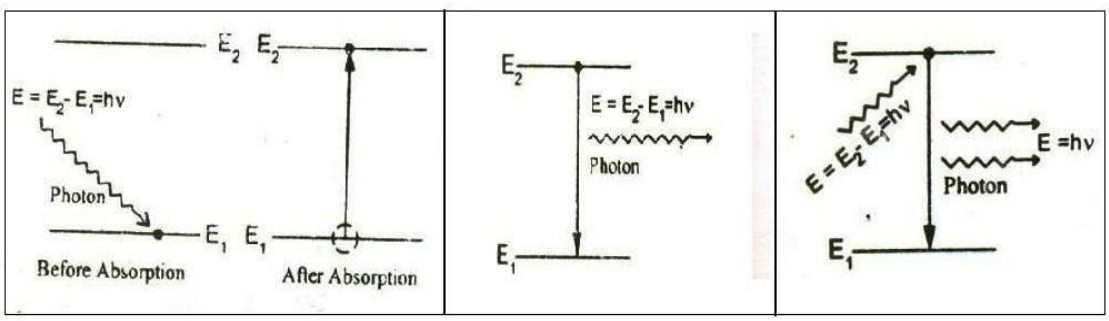


Fig. 1.7 Various process involved in Laser

If there are many numbers of atoms in the ground state then each atom will absorb the energy from the incident photon and goes to the excited state then,

The rate of absorption (R_{12}) is proportional to the following

$$R_{12} \propto \rho_\nu N_1 \qquad \longrightarrow (1)$$

$$R_{12} = B_{12}\rho_\nu N_1 \qquad \longrightarrow (2)$$

Where ρ_ν = Energy density of incident radiation, N_1 = no. of atoms in the ground state and B_{12}- is a constant which gives the probability of absorption transition per unit time.

Normally, the atoms in the excited state will not stay there for a long period of time, rather it comes to ground state by emitting a photon of energy $E = h\nu$. Such an emission takes place by one of the following two methods.

b). Spontaneous emission:

The atom in the excited state returns to the ground state by emitting a photon of energy $E = (E_2 - E_1) = h\nu$ spontaneously without any external triggering as shown in the figure. This process is known as spontaneous emission. Such an emission is random and is independent of incident radiation. If N_1 and N_2 are the numbers of atoms in the ground state (E_1) and excited state (E_2) respectively, then

The rate of spontaneous emission is $R_{21}(Sp) \propto N_2 \quad ---\rightarrow (3)$

$$R_{21}(Sp) = A_{21}\, N_2 \quad ---\rightarrow (4)$$

Where A_{21-} is a constant which gives the probability of spontaneous emission transitions per unit time.

c). Stimulated Emission:

The atom in the excited state can also return to the ground state by external triggering or inducement of photon thereby emitting a photon of energy equal to the energy of the incident photon, known as stimulated emission. Thus results in two photons of same energy, phase difference and of same directionality as shown.

Therefore, the rate of stimulated emission is given by

$$R_{21}(St) \propto \rho_\nu N_2 \quad ---\rightarrow (5)$$

$$R_{21}(St) = B_{21}\rho_\nu N_2 \quad ---\rightarrow (6)$$

Where B_{21-} is a constant which gives the probability of stimulated emission transitions per unit time.

Einstein's theory

Einstein's theory of absorption and emission of light by an atom is based on Planck's theory of radiation. Also under thermal equilibrium, the population of energy levels obeys the Maxwell Boltzmann distribution law

Under thermal equilibrium

$$The\ rate\ of\ Absorption\ =\ The\ rate\ of\ Emission$$

$$B_{12}\rho_\nu N_1 = A_{21}\, N_2 + B_{21}\rho_\nu N_2$$

$$B_{12}\rho_\nu N_1 - B_{21}\rho_\nu N_2 = A_{21}\, N_2 \text{ (or) } \rho_\nu(B_{12}N_1 - B_{21}N_2) = A_{21}\, N_2$$

$$\rho_\nu - \frac{A_{21}\, N_2}{(B_{12}N_1 - B_{21}N_2)}$$

$$\rho_\nu = \frac{A_{21}}{\left(B_{12}\frac{N_1}{N_2} - B_{21}\right)} \quad ---\rightarrow (7)$$

We know from the Boltzmann distribution law

$$N_1 = N_0\, e^{-E_1/K_B T}$$

$$N_2 = N_0 e^{-E_2/K_B T}$$

Where K_B is the Boltzmann Constant, T is the absolute temperature and N_0 is the number of atoms at absolute zero.

At equilibrium, we can write the ratio of population levels as follows

$$\frac{N_1}{N_2} = e^{\frac{E_2 - E_1}{K_B T}}$$

$$\frac{N_1}{N_2} = e^{\frac{h\nu}{K_B T}} \quad ---\rightarrow (8)$$

Substituting equation (8) in equation (9)

$$\rho_\nu = \frac{A_{21}}{\left(B_{12}\left(e^{\frac{h\nu}{K_B T}}\right) - B_{21}\right)}$$

$$\rho_\nu = \frac{A_{21}}{B_{21}} \frac{1}{\left(\frac{B_{12}}{B_{21}}\left(e^{\frac{h\nu}{K_B T}}\right) - 1\right)} \quad ---\rightarrow (10)$$

This equation has a very good agreement with Planck's energy distribution radiation law.

$$\rho_\nu = \frac{8\pi h\nu^3}{C^3} \frac{1}{e^{\frac{h\nu}{K_B T}} - 1} \quad ---\rightarrow (11)$$

Therefore comparing equations (6) and (7) , we can write

$$B_{12} = B_{21} = B \quad \text{and} \quad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{C^3} \quad ---\rightarrow (12)$$

Taking A_{21} =A

The constants ***A*** and ***B*** are called as Einstein Coefficients, which accounts for spontaneous and stimulated emission probabilities.

Generally Spontaneous emission is more predominant in the optical region (Ordinary light). To increase the number of coherent photons stimulated emission should dominate over spontaneous emission. To achieve this, an artificial condition called Population Inversion is necessary.

1.6 PRINCIPLE OF LASER ACTION

Let as consider many number atoms in the excited state. We know the photons emitted during stimulated emission have same frequency, energy and are in phase as the incident photon. Thus result (fig. 1.7) in 2 photons of similar properties.

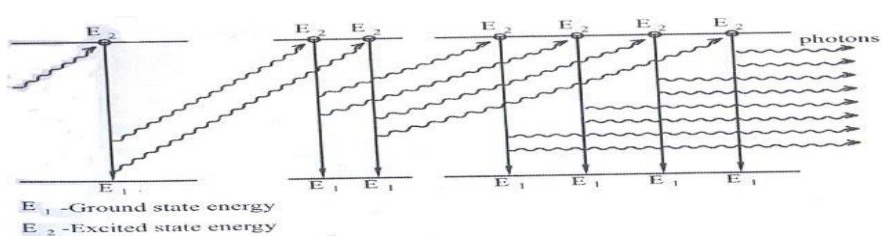


Fig. 1.7 Amplification in Laser process

These two photons induce stimulated emission of 2 atoms in excited state there by resulting in 4 photons. These 4 photons induce 4 more atoms and give rise to 8 photons etc., as shown in figure.

Principle:

Due to stimulated emission the photons multiply in each step-giving rise to an intense beam of photons that are coherent and moving in the same direction. Hence the light is amplified by Stimulated Emission of the Radiation termed LASER.

ACTIVE MEDIUM

A medium in which population inversion can be achieved is known as active medium.

ACTIVE CENTER

The material in which the atoms are raised to the excited state to achieve Population Inversion is called Active Center.

1.7 PUMPING ACTION

The process to achieve the population inversion in the medium is called Pumping action. It is essential requirement for producing a laser beam.

Methods of pumping action

The methods commonly used for pumping action are:

1. Optical pumping (Excitation by Photons)
2. Electrical discharge method (Excitation by electrons)
3. Direct conversion
4. In elastic atom – atom collision between atoms

1. Optical pumping

When the atoms are exposed to light radiations energy, atoms in the lower energy state absorb these radiations and they go to the excited state. This method is called Optical pumping. It is used in solid state lasers like ruby laser and Nd-YAG laser. In ruby laser, xenon flash lamp is used as pumping source.

2. Electrical discharge method (Excitation by electrons)

In this method, the electrons are produced in an electrical discharge tube. These electrons are accelerated to high velocities by a strong electrical field. These accelerated electrons collide with the gas atoms.

By the process, energy from the electrons is transferred to gas atoms. Some atoms gain energy and they go to the excited state. This results in population inversion. This method is called Electrical discharge method. It is represented by the equation

$$A + e^* = A^* + e$$

Where A – gas atom in the ground state A* = same gas atom in the excited state

e* = Electrons with higher Kinetic energy e – Same electron with lesser energy.

This method of pumping is used in gas lasers like argon and CO_2 Laser.

3. Direct Conversion

In this method, due to electrical energy applied in direct band gap semiconductor like Ga As, recombination of electrons and holes takes place. During the recombination process, the electrical energy is directly is converted into light energy.

4. In elastic atom – atom collision

In this method, a combination of two gases (Say A and B are used). The excited states of A and B nearly coincides in energy.

In the first step during the electrical discharge atoms of gas A are excited to their higher energy state A* (metastable state) due to collision with the electrons . A + e* = A* + e

Now A* atoms at higher energy state collide with b atoms in the lower state. Due to inelastic atom - atom collision B atoms gain energy and they are excited to a higher state B*. Hence, A atoms lose energy and return to lower state. A* + B = A + B*

1.8 OPTICAL RESONATOR

An optical resonator consists of a pair of reflecting surfaces in which one is fully reflecting (R_1) and the other is partially reflecting (R_2). The active material is placed in between these two reflecting surfaces.

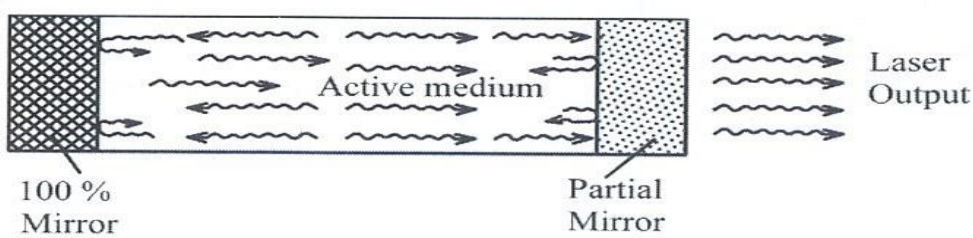


Fig. 1.8 View of optical resonator

The photons generated due to transitions between the energy states of active material are bounced back and forth between two reflecting surfaces. This will induce more and more stimulated transition leading to laser action.

- Interaction of radiation with matter is better explained using concept of photon rather than by the wave concept.
- Energy exchange can take place only at certain discrete values for which the photon energy is the minimum energy unit that light can give or accept.
- Wave picture of light is Classical and Photon picture is Quantum Mechanical.
- Laser- inherently a Quantum Mechanical device------- its operation depends on the existence of photons.
- Maxwell: Light belongs to group of EM waves; propagate with speed "c" in vacuum.
    - ✓ Frequency and wavelength related through

Light incident on a substance, may undergo reflection, transmission, absorption and scattering to varying degrees depending on nature of substance.

- Results in loss of energy and hence

  decrease in light intensity with distance

✓ Absorption or Attenuation

- Attenuation Coefficient (α) - A measure of absorption of light in an optical medium. Is different for different medium and is a function of incident energy.

- At temperature above 0K,

- Atoms always have some thermal energy;

- Distributed among available energy levels according to their energy.

- At Thermal Equilibrium;

- Population at each energy level decreases with increase of energy level,

  For energy levels E_1 and E_2,

- Populations can be computed with Boltzmann's equation

- Ratio of populations, N_2/N_1 is called Relative Population.

- Relative Population (N_2/N_1); dependent on two factors

  - Energy difference (E_2-E_1)

  - Temperature, T

- At Lower Temperature; All atoms are in the ground states.

- At higher Temperature; Atoms move to higher states

- Relative Population (N_2/N_1); dependent on two factors

  - Energy difference (E_2-E_1)

  - Temperature, T

- At Lower Temperature; All atoms are in the ground states.

- At higher Temperature; Atoms move to higher states

<u>Important Conclusions</u>

- As long as the material is in thermal equilibrium, the population of the higher state cannot exceed the population of lower states

- Excitation: Electron in the ground state receives an amount of energy equal to the difference of energy of ground state and one of the excited states, absorbs energy and jumps to the excited state. Electron cannot stay in the excited state for a longer time.

- Has to get rid of the excess energy in order to come to the lower energy level
- ✓ Only mechanism is through emission of a photon.
- ✓ De-excitation: The excited electron emits a photon of energy, $h\nu = (E_2 - E_1)$
- ✓ and jumps from excited state to the ground state ⇒ Spontaneous Emission

1.9 METASTABLE STATE

- ✓ An atom can be excited to a higher level by supplying energy to it. Normally, excited states have short lifetimes ⇒ nanoseconds (10^{-9} s) and release their excess energy by spontaneous emission.
- ✓ Atoms do not stay at such excited states long enough to be stimulated to emit their energy. Though, the pumping agent continuously raises the atoms to the excited level, many of them rapidly undergo spontaneous transitions to the lower energy level Population inversion cannot be established.
- ✓ For establishing population inversion, the excited atoms are required to "wait" at the upper lasing level till a large number of atoms accumulate at that level.
- longer-lived upper levels from where an excited atom does not return to lower level at once, but remains excited for an appreciable time, are known as Metastable States.
- ❖ Atoms stay in metastable states for about 10^{-6} to 10^{-3}s. This is 10^3 to 10^6 times longer than the time of stay at excited levels.
- ❖ Possible for a large number of atoms to accumulate at a metastable level.
- ❖ The metastable state population can exceed the population of a lower level and lead to the state of population inversion.
- ❖ If the metastable states do not exist, What happens ????
  - there could be no population inversion, no stimulated emission and hence no laser operation.

1.10 THRESHOLD CONDITION

- Light bouncing back and forth in the optical resonator
- Undergoes amplification as well as suffers various losses
- Losses occur mainly due to
- (i) Transmission at the output mirror
- (ii) Scattering & Diffraction of light within the active medium.
- For the proper build up of oscillations
- Essential is that the amplification between two consecutive reflections of light from reflecting end mirror can balance losses.

- Determination of threshold gain by considering the change in intensity of a beam of light undergoing a round trip within the resonator ?

  - Consider the laser medium fills the space between the mirrors M_1 & M_2, of reflectivity R_1 & R_2 respectively and mirrors separated by a distance L.

  - Let I_0 - the intensity of the light beam at M_1

- Traveling from mirror M_1 to mirror M_2 ⇒ beam intensity increases from I_0 to I(L),

- After reflection at M_2, the beam Intensity will be;

- After a complete round trip (Reflection from M_1), the final Intensity will be

$$I(2L) = R_1 R_2 I_0 e^{(\gamma - \alpha_s)2L}$$

Growth of output Power Through Cavity

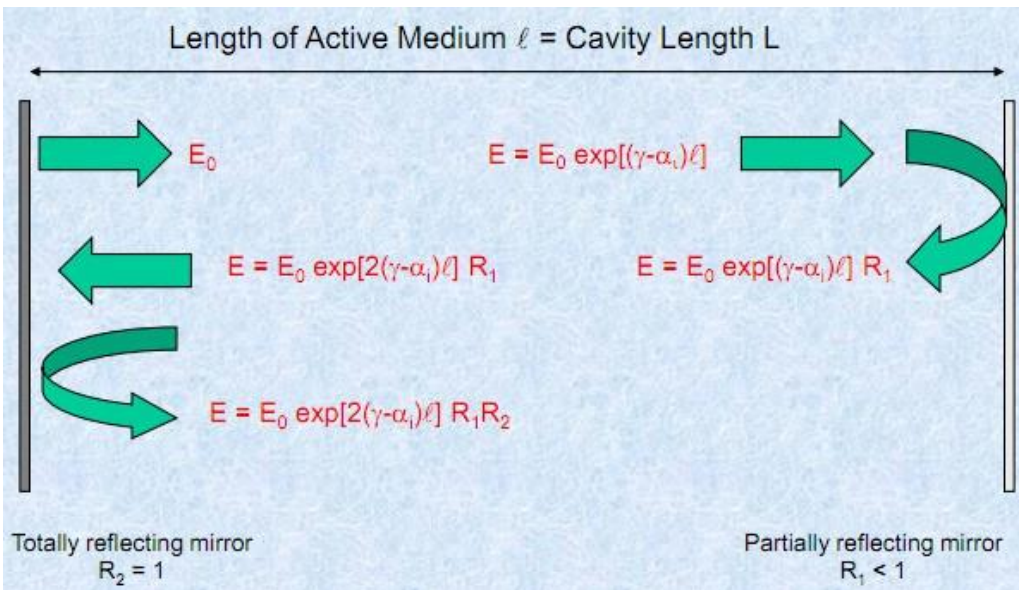


Fig. 1.9 setup of optical resonator

Consider the laser medium fills the space between the mirrors M_1 & M_2, of reflectivity R_1 & R_2 respectively and mirrors separated by a distance L

Let I_0 - the intensity of the light beam at M_1

Let E_0 – the Energy of the light beam at M_1

Product R_1 R_2 represents the losses at the mirrors, whereas α_s includes all the distributed losses such as scattering, diffraction and absorption occurring in the medium.

$$2L(\gamma-\alpha_s) \geq -\ln(R_1 R_2)$$

$$(\gamma-\alpha_s) \geq -\frac{1}{2L}\ln(R_1 R_2)$$

$$\gamma \geq \alpha_s - \frac{1}{2L}\ln(R_1 R_2)$$

$$\gamma \geq \alpha_s + \frac{1}{2L}\ln\left(\frac{1}{R_1 R_2}\right)$$

Condition for Lasing

Shows that the initial gain must exceed the sum of losses in the cavity. The condition is used to determine the threshold value of pumping energy necessary for lasing action.

- 'γ'- Amplification of the laser, dependent on how hard the laser medium is pumped.
- As the pump power is slowly increased, a value of 'γ_{th}' called threshold value will be reached and the laser starts oscillating.

Threshold value 'γ_{th}' is given by

$$\gamma_{th} = \alpha_s + \frac{1}{2L}\ln\left(\frac{1}{R_1 R_2}\right)$$

For the laser to oscillate, $\gamma > \gamma_{th}$ $\Rightarrow$ **Threshold condition for lasing**

- This states the criterion when the net gain would be able to counteract the effect of losses in the cavity

- Value of 'γ' must be atleast 'γ_{th}' for laser oscillations to commence
- If $\gamma > \gamma_{th}$ the waves grow and the amplifier reaches saturation.
- It lowers the value of γ in turn and eventually an equilibrium value is attained at γ_{th}

LINESHAPE FUNCTION

Define lineshape function g(ν)

g(ν) gives the probability that a transition between two levels is an emission (or absorption) of photon whose frequency lies in the range ν and ν+dν . Normalization demands

$$\int_0^\infty g(\nu)d\nu = 1$$

If N is the number of atoms in a given energy level, the spectral distribution of population in the level is given by N(ν) = g(ν)N, i.e., N(ν)dν is the number of atoms in the levels within frequency range ν and ν+d ν, so that

$$\int_0^\infty N(\nu)d\nu = N\int_0^\infty g(\nu)d\nu = N$$

Using the above, one can rewrite the equation defining Einstein's A - coefficient for spontaneous emission

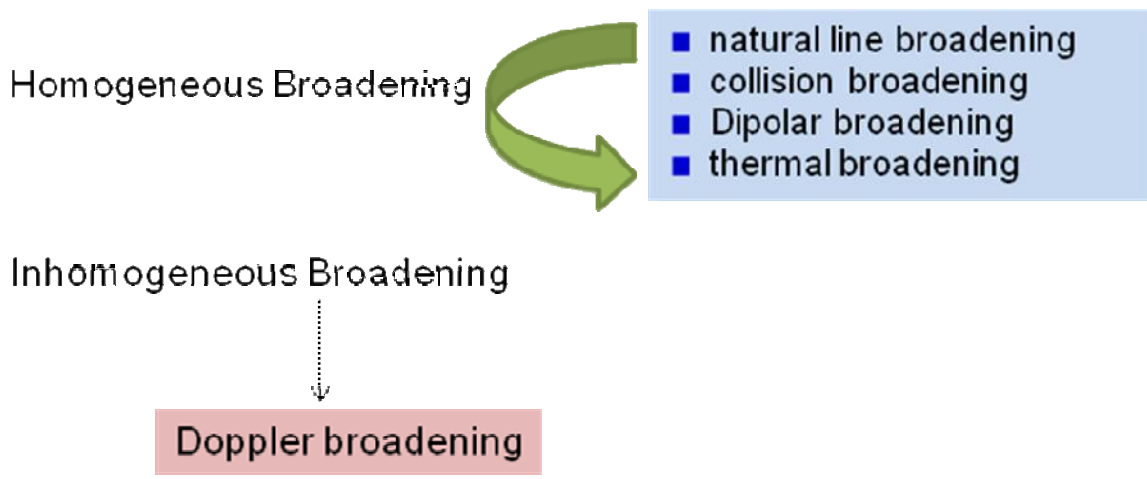
$$\frac{\partial N_2}{\partial t} = -AN_2$$

$$\frac{\partial N_2(\nu)}{\partial t} = -AN_2 g(\nu)\,d\nu$$

g(ν) is a complicated function which depends on the transition involved and also on

external factors.

The broadening of spectral lines is classified into two major categories :



1,11 Febry-Perot resonator

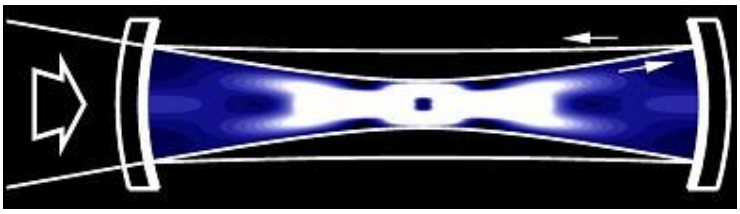


Fig. 1.10 View of Febry-Perot resonator

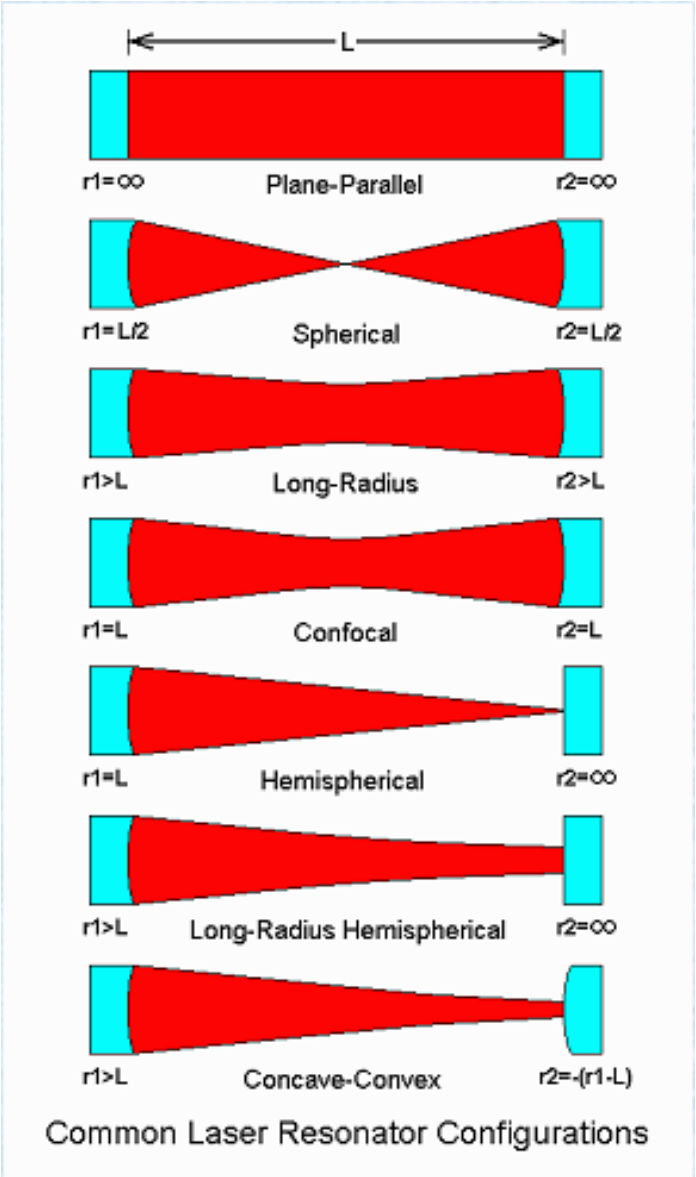


Fig. 1.11 Common model of Laser cavity

1.12 Laser Modes

A wave of frequency ν, that travel along the axis of cavity forms a series of standing waves within the cavity.

- They are discrete resonant conditions determined by the physical dimensions of the cavity.
- Modes governed by the cross-sectional dimension of the optical cavity - Transverse modes
- modes governed by the axial dimension of the resonant cavity - Longitudinal or Axial modes
- In a cavity flanked by two plane parallel mirrors, the standing waves in the cavity satisfy the condition. The axial modes contribute to a single spot of light in the laser spot.

Pumping Schemes

- Atoms characterized by a large number of energy levels.
- Only two, three or four levels are pertinent to the pumping process.

Types are

- Two-level,
- Three-level and
- Four –level schemes.

Two-level Pumping system: Appears to be most simple and straight-forward method to establish population inversion;

- Pumping an excess of atoms into the higher energy state by applying intense radiation fig. 2.2
- A two-level pumping scheme is not suitable for attaining P.I.

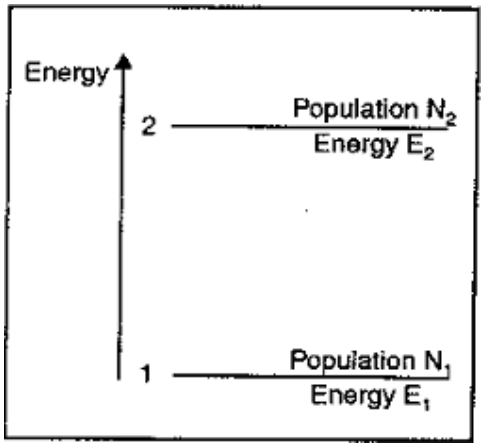


Fig. 1.12 Two level Laser system

P.I. requires the lifetime Δt of upper level E_2 must be longer.

- Achieving population inversion in a two-level atom is not very practical.
- Such a task would require a very strong pumping transition that would send any decaying atom back into its excited state.
- This would be similar to reversing the flow of water in a water fall. It can be done, but is very energy costly and inefficient.
- In a sense, the pumping transition would have to work against the lasing transition.
- That is to say, once the population inversion is achieved the laser would lase.
- But immediately it would end up with more atoms in the lower level.
- Such two-level lasers involve a more complicated process.
- inversion is a familiar prevalent physical system this is not the usual case. Because the probalities for raising an electron to the upper level and inducing the decay of electrons to lower level known as population inversion are exactly the same, so optical pumping will at most only achieve equal population of a two level system.
- In simple words, when both levels are equally populated,the no of electrons going up and down will be same and so the most important ingredient population inversion can not be achieved in case of two levels.
- The only way out is to use a third METASTABLE STATE to solve the problem.

1.13 Three Level Pumping Scheme

A three level scheme; Lower level is either the ground state or a level whose separation from the ground state is small compared to kT as shown in fig. 2.3.

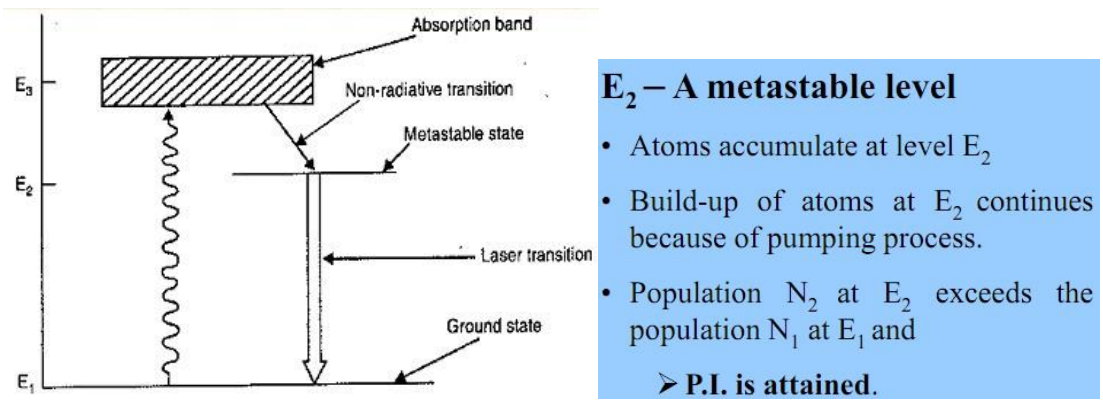


Fig. 1.13 Three level Laser system

❖ A photon of $h\nu(=E_2-E_1)$ can induce stimulated emission and laser action.

Major disadvantage of a three level scheme ⇒ it requires very high pump powers.

- Terminal level of the laser transition is the ground state.

- As the ground state is heavily populated, large pumping power is to be used to depopulate the ground level to the required extent ($N_2 > N_1$)

Three level scheme can produce light only in Pulses.

- Once stimulated emission commences, the metastable state E_2 gets depopulated very rapidly and the population of the ground state increases quickly.
- As a result the population inversion ends. One has to wait till the population inversion is again established.
- Three level lasers operate in Pulsed Mode.

1.14 Four Level Pumping Scheme

E_3 - a metastable level

- Laser transition takes the atoms to the level E2
- Atoms lose the rest of their excess energy & finally reach the ground state E1.
- Atoms are once again available for excitation.

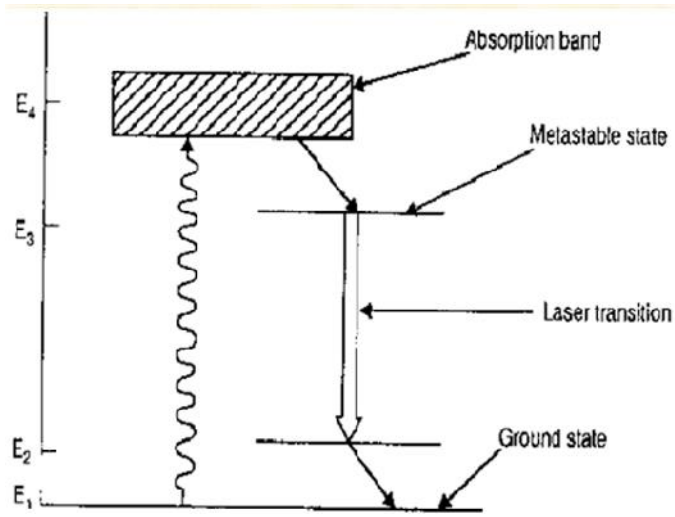


Fig. 1.14 Four level Laser System

In Four level scheme, the terminal laser level E_2 is well above the ground level such that $(E_2-E_1) >> kT$.

- It guarantees that the thermal equilibrium population of E_2 level is negligible.
- In contrast to three level scheme, the lower laser transition level in four level scheme is not the ground state and is virtually vacant.
- It requires less pumping energy than does a three level laser.
- Further, the lifetime of the lower laser transition level E_2 is much shorter, hence atoms in level E_2 quickly drop to the ground state.
- This steady depletion of E_2 level helps sustain the population inversion by avoiding an accumulation of atoms in the lower lasing level fig. 2.4.
- Four level lasers can operate in Continuous Wave mode
- Most of the working lasers are based on Four Level Scheme

Comparison of

Three level and

Four level

Systems Three

level laser,

$N_{th} = (N_2 - N_1)$ and $N_0 = N_2 + N_1$

For the laser to begin lasing ; $N_2 > \frac{N_0}{2} + \frac{N_{th}}{2}$

As $N_0 >> N_{th}$, therefore, $N_2 > \frac{N_0}{2}$

Four level laser

$$N_2 = N_1 \exp\left(\frac{(E_2 - E_1)}{kT}\right)$$

For the laser to begin lasing; $(N_3 - N_2) > N_{th}$ i.e. $N_3 > N_{th}$

$$\therefore \quad \frac{(N_{th})_{3-level}}{(N_{th})_{4-level}} = \frac{N_0}{2N_{th}} \Rightarrow \textbf{a very large quantity}$$

Implies that it is much easier to pump a four level laser than a three level laser. This is the reason why most of the lasers are of four-level.